

# NEXAR: Intelligent Quantum-Classical Workload Routing Platform

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**Abstract**—Determining whether a workload benefits from quantum or classical execution remains a key open challenge in the NISQ era. We present Nexar, a routing platform that analyzes source code, automatically extracts workload features, and recommends an execution target that jointly optimizes cost and runtime. Nexar pairs a physics-aware rule engine for clear-cut cases: workloads with no quantum structure, and workloads too large for classical simulation, with a weighted ensemble of machine learning, rules, and cost analysis that handles the ambiguous mid-scale regime where rules defer. Feature extraction is automated using CodeBERT-based algorithm classification and quantum circuit characterization. Across a benchmark spanning search, factorization, optimization, and simulation workloads, Nexar achieves meaningful agreement with independently-grounded oracles for algorithmic advantage and hardware viability, substantially outperforming random, cost-only, and rule-only baselines on the Quantum-class recovery metrics that matter for this imbalanced benchmark. Routing decisions are further validated end-to-end on IBM superconducting hardware, including a large-qubit variational circuit, demonstrating that Nexar’s recommendations execute successfully on production quantum devices even where noise dominates output fidelity absent error mitigation.

**Index Terms**—CodeBERT, cost analysis, decision engine, hybrid computing, NISQ, quantum advantage, quantum computing, workload routing

## I. INTRODUCTION

The advent of Noisy Intermediate-Scale Quantum (NISQ) devices [1] has created a computational resource-allocation challenge: given a workload, should it execute on classical hardware, a quantum simulator, or a real quantum device? The decision depends on a complex interplay of problem structure, circuit complexity, hardware noise, cost, and qubit availability. Classical state-vector simulation remains feasible

up to approximately 50 qubits, beyond which memory requirements grow as  $O(2^n)$  and exceed any deployable system, creating a crossover where full state-vector simulation of high-entanglement circuits becomes infeasible. Yet simply having a quantum device does not guarantee advantage: transpilation, shots, queue times, and error rates mean many polynomial-solvable problems remain orders of magnitude faster and cheaper on classical hardware. The challenge is not whether quantum computing will eventually provide advantage, but when and for which problems it does so today.

This paper presents Nexar, a full-stack platform that automates the analysis, classification, and routing of computational workloads across quantum and classical execution backends. Its contributions are:

- 1) **Automated Code Analysis Pipeline:** A multi-dimensional analysis engine that detects programming languages across five quantum and classical frameworks, computes complexity metrics, and classifies algorithms using a three-tier approach combining CodeBERT transformers, ensemble machine learning, and rule-based pattern matching.
- 2) **Two-Stage Decision Engine:** A physics-aware rule engine that deterministically resolves clear-cut workloads, coupled with a weighted ML+rules+cost ensemble (weights 50%/35%/15%) that handles the NISQ-ambiguous regime where rules defer.
- 3) **Provider-Extensible Execution Layer:** A hardware abstraction layer with a unified API; the IBM Qiskit backend is fully implemented and used for all real-hardware validation, while AWS Braket and Azure

Quantum provider classes scaffold future cross-backend evaluation.

- 4) **Empirical Evaluation:** A comprehensive benchmark across 100 workloads spanning classical, hybrid, and pure quantum circuits, with cost and performance analysis demonstrating routing accuracy and the current state of quantum-classical cost tradeoffs.

## II. RELATED WORK

### A. Quantum-Classical Hybrid Frameworks

Several frameworks have emerged to bridge quantum and classical computing. Weder et al. [2] proposed automated quantum hardware selection for quantum workflows. Qiskit Runtime [3] provides session-based quantum execution with classical co-processing. Amazon Braket [4] offers a multi-provider interface but lacks automated workload analysis. Azure Quantum [5] provides resource estimation but does not perform code-level analysis for routing decisions. Tang et al. [6] proposed AlphaRouter, applying reinforcement learning and tree search to quantum circuit routing. Zhan et al. [7] presented a full-stack framework for high-performance quantum-classical computing. Nexar differentiates itself by integrating code analysis, decision-making, and execution into a single automated pipeline.

*Positioning for empirical comparison.* These systems operate at different layers of the quantum-classical stack and do not admit a direct head-to-head comparison: AlphaRouter [6] maps physical qubits for circuits *already* assumed quantum; execution platforms [3]–[5] accept whatever the user submits; and Weder et al. [2] selects among quantum backends for pre-classified quantum workflows. None answers Nexar’s core question: “given arbitrary source code, should this run quantum or classical?”, so our evaluation (Section VIII-B) compares against a qubit-threshold heuristic, a circuit-volume heuristic, a 1,000-seed random router, and Nexar’s own components in isolation. The rules-only ablation subsumes the Weder-style rule-based selection logic applied to our 16-feature vector.

### B. Quantum Resource Estimation

Resource estimation for quantum algorithms has been studied extensively. Lammers et al. [8] addressed quantum resource management challenges in the NISQ era [9]. Beverland et al. [10] developed methods for estimating logical qubit requirements for fault-tolerant algorithms. Quetschlich et al. [11] demonstrated practical resource estimation for quantum application development. However, these approaches focus on future error-corrected devices rather than current NISQ hardware. Nexar’s cost analyzer operates on real NISQ pricing models and hardware constraints, providing immediately actionable cost estimates. Recent work by Kashif et al. [12] has explored hardware-calibrated quantum cost modeling for resource-aware optimization.

### C. Algorithm Classification and Code Analysis

Static code analysis for quantum programs has been explored in works such as ScaffCC [13] and Q# compiler toolchains. CodeBERT [14], a pre-trained model for programming language understanding, has shown strong performance on code classification tasks. Afrose et al. [15] further applied CodeBERT for translating QASM to QIR. Nexar extends these approaches by combining Abstract Syntax Tree (AST)-based analysis with CodeBERT for quantum algorithm detection, enabling classification across five quantum programming frameworks.

### D. Quantum Advantage Thresholds

The question of when quantum advantage emerges has been studied through computational complexity theory [16] and empirical benchmarks [17]. Our work contributes empirical data on the physical-feasibility crossover: 50–60 qubits is where classical state-vector memory (16 PB to 16 EB) exceeds any deployable system, making full state-vector simulation of high-entanglement circuits infeasible beyond this scale.

## III. SYSTEM ARCHITECTURE

Nexar is a six-service microservices monorepo coordinated by a central API Gateway (Fig. 1).

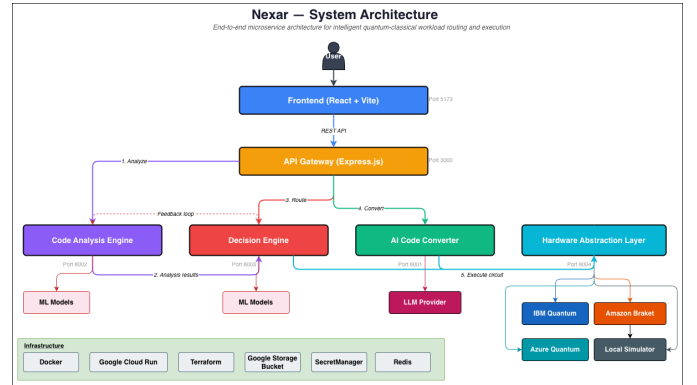


Fig. 1. Nexar architecture: source code flows from Frontend to Code Analysis, Decision Engine, AI Code Converter, and Hardware Abstraction Layer through the API Gateway.

Submitted code enters a JWT-authenticated Express.js API Gateway, then the Code Analysis Engine extracts language, complexity, and algorithm signals. The Decision Engine recommends a target backend; when classical code is routed to quantum execution, the AI Code Converter generates Qiskit before the Hardware Abstraction Layer dispatches the job. Services are containerized with Docker and deployed on Google Cloud Run via Terraform.

## IV. CODE ANALYSIS ENGINE

The Code Analysis Engine transforms raw source code into the structured feature set downstream components consume (Fig. 2).

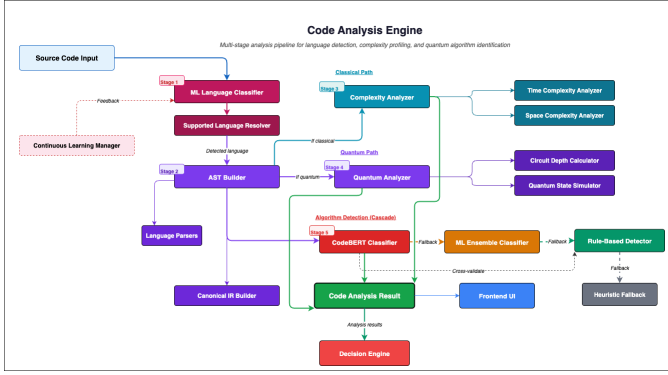


Fig. 2. Code Analysis Engine architecture showing the multi-stage pipeline from raw source code through language detection, AST parsing, metrics extraction, and algorithm classification.

### A. Language Detection

The engine supports five languages: Python, Qiskit, Cirq, Q#, and OpenQASM. Detection uses a four-model soft-voting ensemble (CodeBERT 0.30, XGBoost 0.20, Random Forest 0.25, Gradient Boosting 0.25), trained on  $\sim 5,400$  files and evaluated on 1,080 held-out samples. The ensemble achieves 98.52% accuracy and macro F1 0.986 (Tables I, II); the only measurable confusion is symmetric between Qiskit and Cirq ( $\approx 2\%$ ), expected given their shared Python host, and no class falls below 0.97 F1.

TABLE I  
LANGUAGE-DETECTION ENSEMBLE: PER-CLASS METRICS ON 1,080  
HELD-OUT TEST SAMPLES

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Cirq                 | 0.985     | 0.978  | 0.982 | 273     |
| OpenQASM             | 1.000     | 0.987  | 0.993 | 229     |
| Python               | 0.976     | 1.000  | 0.988 | 206     |
| Qiskit               | 0.968     | 0.968  | 0.968 | 220     |
| Q#                   | 1.000     | 1.000  | 1.000 | 152     |
| <b>Macro avg.</b>    | 0.986     | 0.987  | 0.986 | 1080    |
| <b>Weighted avg.</b> | 0.985     | 0.985  | 0.985 | 1080    |

TABLE II  
LANGUAGE-DETECTION CONFUSION MATRIX (ROWS = TRUE, COLUMNS  
= PREDICTED)

|          | Cirq       | OpenQASM   | Python     | Qiskit     | Q#         |
|----------|------------|------------|------------|------------|------------|
| Cirq     | <b>267</b> | 0          | 2          | 4          | 0          |
| OpenQASM | 0          | <b>226</b> | 0          | 3          | 0          |
| Python   | 0          | 0          | <b>206</b> | 0          | 0          |
| Qiskit   | 4          | 0          | 3          | <b>213</b> | 0          |
| Q#       | 0          | 0          | 0          | 0          | <b>152</b> |

### B. Metrics Extraction

For all detected code, the engine computes classical software metrics via tree-sitter-based AST parsing, including cyclomatic complexity [18], cognitive complexity, time and space complexity classification (from  $O(1)$  to  $O(2^n)$ , where

$n$  denotes the input size), and structural metrics (loop count, nesting depth). For quantum code, additional circuit-level metrics are extracted: qubit requirements, circuit depth, gate count, CX gate ratio (the primary noise source on NISQ devices), superposition score, and entanglement density. The complete feature set is summarized in Table III.

TABLE III  
RAW CODE ANALYSIS FEATURES

| Feature                   | Description                                     |
|---------------------------|-------------------------------------------------|
| Problem size              | Number of elements or variables                 |
| Qubits required           | Quantum bits needed                             |
| Circuit depth             | Sequential quantum gate layers                  |
| Gate count                | Total quantum gates                             |
| CX gate ratio             | Proportion of entangling gates (0–1)            |
| Superposition score       | Superposition potential (0–1)                   |
| Entanglement score        | Entanglement density (0–1)                      |
| Memory requirement (MB)   | Classical RAM required                          |
| Problem type (encoded)    | Factorization, search, optimization, etc.       |
| Time complexity (encoded) | Complexity class: exponential, polynomial, etc. |

### C. Algorithm Classification

Algorithm detection employs a three-tier cascade: (1) *CodeBERT Transformer* [14] classifies the algorithm family from code semantics; (2) *ML Ensemble* (Random Forest, XGBoost, Gradient Boosting) uses 35 hand-engineered circuit features; (3) *Rule-Based Pattern Matching* identifies canonical structures (QFT, Grover oracles, VQE ansatz). The system selects the highest-confidence result and falls back through tiers below threshold; tier-3 additionally overrides tier-1 when rule confidence  $\geq 0.7$  disagrees with CodeBERT’s top prediction. Tier-1 CodeBERT attains macro F1 0.66 across eight families (750-sample held-out), perfect on Grover/QAOA/QPE/Shor but missing Bernstein–Vazirani and Deutsch–Jozsa (F1=0) whose Python syntax lacks signal for a general-code transformer. Tier-2 recovers these at macro F1 0.921 (acc. 0.864,  $n=8,800$ ) via CX-gate patterns invisible to CodeBERT.

*Tier fire-rate on the real benchmark.*: Over the 100-workload benchmark, 33 classical-language inputs skip the quantum pipeline; of the 67 quantum workloads, tier-1 wins 10 (14.9%), tier-3 carries 57 (85.1%; 46 cross-validation overrides + 11 low-confidence fallbacks), and tier-2 never fires; CodeBERT’s internal fallback always populates top-1, so tier-2 is a reserve for CodeBERT unavailability rather than a routine contributor. Tier-1 thus handles canonical textbook circuits; tier-3 handles everything else.

## V. DECISION ENGINE

The Decision Engine is Nexar’s core component, implementing a hybrid prediction system that combines three sub-components through weighted voting (Fig. 3).

### A. Feature Vector Design

The engine operates on a 16-feature input vector composed of 10 raw features (Table III) and 6 derived physics features (Table IV):

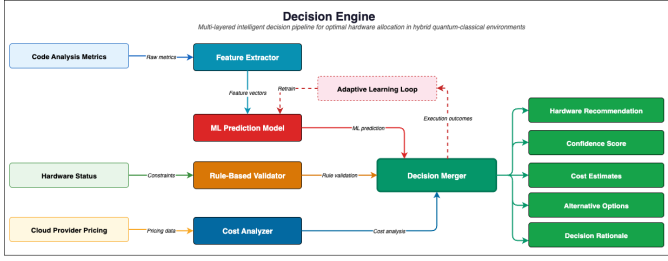


Fig. 3. Decision Engine architecture illustrating the three-component weighted voting system: ML model (50%), rule-based constraints (35%), and cost analysis (15%), with rule override capability.

TABLE IV  
DERIVED PHYSICS FEATURES

| Feature                | Formula                                                    | Purpose                          |
|------------------------|------------------------------------------------------------|----------------------------------|
| Circuit Volume         | $\text{Qubits} \times \text{Depth}$                        | Quantum resource intensity       |
| Noise Sensitivity      | $\text{Qubits} \times \text{Depth} \times \text{CX Ratio}$ | Error accumulation prediction    |
| Quantum Overhead Ratio | $\text{Problem Size} / \text{Qubits}$                      | Encoding efficiency              |
| NISQ Viability Score   | Composite penalty function                                 | Feasibility on current hardware  |
| Gate Density           | $\text{Gate Count} / \text{Qubits}$                        | Operational complexity per qubit |
| Entanglement Factor    | $\text{CX Ratio} \times \text{Entanglement}$               | Combined entanglement potential  |

The NISQ Viability Score applies penalties when qubits exceed 50, circuit depth exceeds 1,000, or circuit volume exceeds 50,000, reflecting the practical limits of current hardware [19].

### B. ML Model Component (50% Weight)

The ML component is a CalibratedClassifierCV (Platt-scaled [20]) wrapping a Random Forest, performing binary classification (Quantum/Classical) over the 16-feature vector. The synthetic training corpus is produced by AST-level mutation of base classical-to-quantum algorithm pairs (factorization, search, period-finding, random walk) with transforms spanning qubit scaling, depth injection, gate-set substitution, and register re-indexing; features are z-score standardized and the model is trained with stratified cross-validation.

On an 876-instance stratified held-out split (438/438), the Random Forest head attains accuracy 0.977,  $F1_{(Q)}$  0.978, ROC-AUC 0.999, Brier 0.016; Gradient Boosting (0.983) and XGBoost (0.984) match within 1 point, confirming the signal is not architecture-specific. Per-problem-type slice accuracy (Table V) floors at 0.941. These synthetic metrics do not transfer: the isolated ML ablation attains only  $F1_{(Q)}=0.424$  on the real oracle evaluation (Table IX), motivating the two-stage pipeline (Sections VIII-B, VIII-B6).

### C. Rule-Based System (35% Weight)

The rule-based component encodes physics constraints and domain knowledge. Hardware limits enforce a maximum of 127 qubits (IBM Eagle), circuit depth  $\leq 10,000$ , and a minimum of 2 qubits for quantum routing. Problem-type rules

TABLE V  
ML CLASSIFIER HELD-OUT ACCURACY BY PROBLEM TYPE (RANDOM FOREST HEAD; STRATIFIED 876-INSTANCE TEST SET).

| Problem Type        | Accuracy     | n          |
|---------------------|--------------|------------|
| dynamic_programming | 1.000        | 71         |
| factorization       | 0.988        | 84         |
| matrix_ops          | 1.000        | 74         |
| optimization        | 0.948        | 134        |
| random_circuit      | 1.000        | 259        |
| search              | 0.941        | 118        |
| simulation          | 0.953        | 107        |
| sorting             | 1.000        | 29         |
| <b>Overall</b>      | <b>0.977</b> | <b>876</b> |

Slice-level floor (search: 0.941) exceeds the aggregate accuracy of every simpler real-benchmark baseline in Table IX, confirming the synthetic held-out difficulty is low relative to the real benchmark and underlines the syn-to-real gap discussed in Section VIII-B6.

favor quantum execution for factorization (Shor’s), search (Grover’s), simulation, and optimization (Quantum Approximate Optimization Algorithm (QAOA) [21]/VQE), while routing sorting, dynamic programming, and matrix operations classically. Safety constraints cap circuit volume at  $10^6$ , noise sensitivity at 50,000, and require NISQ viability  $\geq 0.1$ . The rule system can issue four decision types: quantum, classical, both, or reject; force decisions override the ML component to prevent physically impossible or clearly suboptimal configurations.

### D. Cost Analysis Component (15% Weight)

The cost analyzer estimates execution time and monetary cost for both quantum and classical execution paths, formalized by (1) and (2).

Gate timing is anchored to calibration data retrieved live via QiskitRuntimeService. Table VI summarises the median metrics across three IBM backends (April 2026); the 68.0 ns median ECR gate time is consistent across devices and directly informs  $t_q$  in (1). The rate  $r_{\text{QPU}} = \$1.60/\text{s}$  [28] is transient, so we report QPU-seconds alongside dollar estimates throughout.

TABLE VI  
IBM QUANTUM BACKEND CALIBRATION DATA (RETRIEVED APRIL 2026 VIA QISKITRUNTIMESERVICE)

| Backend       | Qubits | 2Q Err. | 1Q Err.  | RO Err. | T1 ( $\mu\text{s}$ ) | T2 ( $\mu\text{s}$ ) |
|---------------|--------|---------|----------|---------|----------------------|----------------------|
| ibm_fez       | 156    | 0.00265 | 0.000316 | 0.01489 | 155.2                | 104.0                |
| ibm_marrakesh | 156    | 0.00251 | 0.000304 | 0.01044 | 184.8                | 98.4                 |
| ibm_kingston  | 156    | 0.00183 | 0.000229 | 0.00867 | 275.4                | 144.8                |
| <i>Median</i> |        | 0.00251 | 0.000304 | 0.01044 | 184.8                | 104.0                |

2Q gate time: 68.0 ns (ECR, consistent across all backends). RO = readout.

Quantum cost (IBM pay-as-you-go [28]):

$$C_q = t_q r_{\text{QPU}}, \quad (1)$$

where  $t_q$  is the QPU reservation time (full billable window including gate execution, readout, reset, and post-processing

across  $n_{\text{shots}}=1024$  shots) and  $r_{\text{QPU}}=\$1.60/\text{s}$ . IBM bills purely on QPU-seconds; AWS Braket’s per-shot/per-task fees would add terms to (1), but our real-hardware validation targets IBM so (1) is the appropriate reference.

Classical cost (AWS EC2 on-demand [29]):

$$C_c = t_c r_{\text{compute}} + M_{\text{GB}} t_c r_{\text{memory}} + C_{\text{overhead}}, \quad (2)$$

where  $t_c$  is classical execution time,  $r_{\text{compute}}=\$0.0001/\text{s}$  ( $\sim \text{c6i.2xlarge}$ ),  $M_{\text{GB}}$  is memory usage,  $r_{\text{memory}}=\$0.00001/(\text{GB}\cdot\text{s})$ , and  $C_{\text{overhead}}=\$0.001$  covers job submission.

### E. Decision Merging

The engine is a two-stage pipeline. In stage one the rule engine emits `FORCE_QUANTUM`, `FORCE_CLASSICAL`, or `REJECT` (deterministic path) or `ALLOW_BOTH` (defers to stage two). Stage two combines the three components by weighted voting,

$$S_{\text{quantum}} = w_{\text{ML}} P_{\text{ML}}^q + w_{\text{rules}} P_{\text{rules}}^q + w_{\text{cost}} P_{\text{cost}}^q, \quad (3)$$

with  $w_{\text{ML}}=0.50$ ,  $w_{\text{rules}}=0.35$ ,  $w_{\text{cost}}=0.15$ .

On our 100-workload benchmark the first stage resolves 61 workloads (40 `FORCE_CLASSICAL`, 20 `FORCE_QUANTUM`, 1 `REJECT`); the merger fires only on the 39 remaining `ALLOW_BOTH` workloads, where  $P_{\text{rules}}^q$  is neutral by construction, so the weighted sum is dominated by the ML probability with cost as a tie-breaker (characterised in Section VIII-B6).

### F. Feature Importance

Random Forest Gini importance shows the ensemble’s decisions are driven by physically meaningful features: `qubits_required` (0.31) and `circuit_volume` (0.24) dominate, followed by `time_complexity_encoded` (0.15), `entanglement_factor` (0.11), and `cx_gate_ratio` (0.08); remaining features contribute  $\leq 0.04$  each. The ranking diverges from pure Pearson correlation against the label (feature\_target\_correlation.csv in the artifact bundle), which places `superposition_score` (0.483) and the entanglement features (0.408–0.435) ahead of `qubits_required` (0.209). This is expected: Gini rewards non-linear decision thresholds (e.g. the 50-qubit simulation wall) and distributes credit across the collinear entanglement triple, while correlation rewards monotonic linear dependence. The Gini ranking reflects what the ensemble uses; the correlation ranking reflects univariate label information.

### G. Weight Sensitivity Analysis

The weights  $(w_{\text{ML}}, w_{\text{rules}}, w_{\text{cost}}) = (0.50, 0.35, 0.15)$  are chosen architecturally, not tuned. The ordering enforces a “no-dictator” property: no single component overrides a decision alone, and any two form a strict majority. The cost weight is deliberately small because it is anchored to a transient pricing epoch (IBM pay-as-you-go, April 2026; Table VI).

TABLE VII  
WEIGHT SENSITIVITY: ROUTING COUNTS ACROSS SIX CONFIGURATIONS ( $N=100$ ).

| Weights<br>(ML / Rules / Cost) | Quantum<br>Routed | Classical*<br>Routed | Changed<br>vs. Base | Stable<br>(%) |
|--------------------------------|-------------------|----------------------|---------------------|---------------|
| <b>50 / 35 / 15</b> (baseline) | <b>20</b>         | <b>80</b>            | —                   | —             |
| 70 / 20 / 10                   | 20                | 80                   | 0                   | 100           |
| 30 / 55 / 15                   | 20                | 80                   | 0                   | 100           |
| 40 / 30 / 30                   | 20                | 80                   | 0                   | 100           |
| 100 / 0 / 0                    | 18                | 82                   | 2                   | 98            |
| 33 / 33 / 34                   | 46                | 54                   | 26                  | 74            |

\*Includes one rule-override Reject (150q oversize), weight-invariant.

Table VII reports routing counts across six configurations. Among the 39 workloads reaching the weighted stage, routing is identical to the baseline under ML-heavy, rule-heavy, and cost-sensitive configurations and flips only two under ML-only. Only the adversarial equal-weighting (33/33/34) produces substantively different routing (26 flips), all on 6–16 qubit workloads well inside the sub-50-qubit regime where classical execution is  $\geq 375\times$  cheaper (Section VIII-D); five of the 26 produced 0.39–5.18% fidelity on `ibm_fez` (Table XV), empirically confirming the baseline as the correct routing target.

## VI. AI CODE CONVERTER

The AI Code Converter translates classical Python into Qiskit circuits when quantum execution is recommended (Fig. 4).

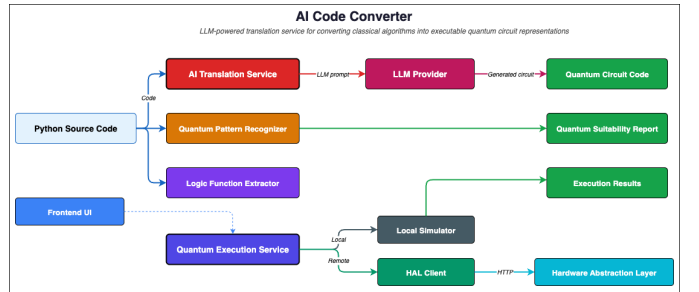


Fig. 4. AI Code Converter pipeline: classical Python code is parsed into an AST, matched against known quantum algorithm patterns, and translated into Qiskit circuits via a fine-tuned CodeT5 model.

### A. Translation Pipeline

The converter uses a fine-tuned CodeT5-base [22] (6+6 layers,  $d_{\text{model}}=512$ , 8 heads) trained on 1,500 curated Python-to-Qiskit pairs covering six algorithm families. The pipeline: (i) *logic extraction* parses the input AST and strips I/O boilerplate; (ii) *pattern recognition* matches extracted logic to templates (e.g. linear search  $\rightarrow$  Grover, parity check  $\rightarrow$  Deutsch, hidden bitstring  $\rightarrow$  Bernstein–Vazirani); (iii) *circuit generation* tokenises at 256 tokens and decodes with beam search ( $k=3$ ) up to 300 output tokens, emitting Qiskit gates, measurements, and classical registers.



## B. Suitability Scoring

Each translation candidate receives a three-tier suitability score (HIGH, MEDIUM, LOW) with associated confidence and estimated speedup. Speedup estimates are derived from theoretical complexity comparisons ( $O(N)$  classical versus  $O(\sqrt{N})$  quantum for Grover [23], for example, where  $N$  is the search-space size), combined with simulator resource metrics from Qiskit Aer execution. Confidence reflects the AST-level overlap between the submitted code and the matched pattern signature, allowing users to judge whether quantum translation offers a meaningful advantage for their specific use case.

## C. Evaluation

The translator was fine-tuned for 7 epochs on the 1,500-example Python-to-Qiskit corpus (final training loss 0.5356, validation loss below 0.01 by epoch 2) and evaluated on a 300-sample held-out test set across six algorithm families. Table VIII reports the key metrics: syntactic validity 98.7%, algorithm-family classification 77.7%, BLEU 28.36, exact-match 0.197. We treat algorithm-family agreement as the primary correctness signal, since it measures whether the translation preserves the intended algorithmic structure independent of gate-ordering choice; EM is reported for completeness but is brittle for code generation (commuting single-qubit gates, qubit-index relabellings, and equivalent decompositions of controlled rotations all produce distinct token sequences), and we do not claim  $EM = 0.197$  as evidence of pervasive semantic equivalence without an output-distribution-level probe. A functional-equivalence audit (simulating predicted and reference circuits on matched inputs under Qiskit Aer and comparing by total variation or fidelity) is deferred as a remaining validation gap (Section IX-B).

TABLE VIII  
AI CODE CONVERTER: EVALUATION ON 300 HELD-OUT SAMPLES (6 ALGORITHM FAMILIES)

| Metric                            | Value        |
|-----------------------------------|--------------|
| Valid Qiskit Circuits (syntactic) | <b>0.987</b> |
| Correct Algorithm Classification  | 0.777        |
| BLEU Score                        | 28.36        |
| Exact Match (strict token-level)  | 0.197        |

## VII. HARDWARE ABSTRACTION LAYER

The Hardware Abstraction Layer provides a provider-extensible interface for quantum job submission. IBM Qiskit is the fully implemented backend used for all real-hardware results in Table XV; AWS Braket and Azure Quantum provider classes implement the same signature as architectural scaffolding but their job-execution methods are stubs in the present release (Section X). The IBM implementation manages the full job lifecycle (device discovery, submission with target device/shots/transpilation options, status tracking for queued/running/completed/failed, and result retrieval with measurement distributions/metrics/metadata) through a

factory-based abstraction that isolates routing from provider-specific SDKs, allowing new providers to be added by implementing a single interface.

## VIII. EXPERIMENTAL EVALUATION

### A. Benchmark Design

We evaluated Nexar on 100 workloads in three categories: *Classical* (32; deep learning, scientific computing, ML, cryptography, and combinatorial optimization), *Hybrid* (23; VQE, QAOA, quantum ML, and ZNE), and *Quantum* (45; 1–150q molecular simulation [25], Shor factoring, Grover search, QAOA optimization, and foundational circuits DJ/BV/Simon/QFT/QPE).

### B. Routing Accuracy

1) *Routing Distribution*: Of the 100 workloads, Nexar routes 79 to classical, 20 to quantum, and rejects one 150-qubit circuit whose full state-vector simulation is infeasible for high-entanglement settings ( $\sim 10^{31}$  MB) and exceeds the 127-qubit IBM Eagle limit. Quantum routes span 20–120 qubits (confidence 0.54–1.0); every 0-qubit workload is routed classically at confidence 1.0.

2) *Dual-Oracle Ground Truth*: Because routing can be judged by different criteria, we evaluate against two independent oracles derived from workload metadata and physical circuit features (qubits, depth, entanglement), without using Nexar outputs. Oracle A is fully independent of Nexar logic. Oracle B uses externally sourced NISQ-viability thresholds (IBM specs + prior error-budget analysis), sharing first-principles physics with the rule engine but no internal co-calibration (Section IX-A).

*Oracle A (Algorithmic Advantage)* labels Quantum when a known quantum algorithm provides theoretical speedup: classical simulation infeasible ( $\geq 50$  qubits; [1]), Shor-class factorization ( $\geq 8$  qubits; [24]), large entangled VQE ( $\geq 20$  qubits, entanglement  $\geq 0.3$ ; [25]), large QAOA ( $\geq 20$  qubits, entanglement  $\geq 0.3$ ; [21]), or Grover search ( $\geq 20$  qubits; [23]); rejects circuits above 127 qubits [17].

*Oracle B (Hardware Viability)* labels Quantum only when current NISQ hardware can produce usable signal. Its depth wall is derived from external pre-Nexar sources: IBM Heron r2  $T_2 \approx 144 \mu s$  and ECR  $t_{\text{gate}} \approx 68 \text{ ns}$  combined with Preskill’s error-budget framework [1],  $d_{\text{max}} \approx T_2 / (K \cdot t_{\text{gate}})$  with  $K=10$ , yielding  $\sim 212$  rounded to a conservative 200 gates. The threshold was fixed before examining any router’s behaviour and is not back-calibrated. Rules: post-transpile depth  $> 200$  at sub-50 qubits is noise-dominated (Classical); sub-20-qubit circuits are trivially simulable (Classical); the sub-50-qubit band with depth  $\leq 200$  and entanglement  $\geq 0.3$  is the practical NISQ-advantage regime. Memory-wall and rejection thresholds match Oracle A.

Under Oracle A, 100 workloads divide 85/14/1 (Classical/Quantum/Reject); under Oracle B, 92/7/1. Every label is reproduced in the supplementary `oracle_ground_truth.csv`.

3) *Baselines and Ablations*: We compare Nexar against three baselines and three component ablations: *B1*, a single-line heuristic routing quantum iff qubits  $\geq 50$ ; *B2*, a circuit-volume heuristic routing quantum iff qubits  $\times$  depth  $\geq 50,000$ ; *B3*, a 50/50 random router averaged over 1,000 seeds; *A1*, the ML component in isolation (Random Forest output with no rule or cost influence); *A2*, the rule-based component in isolation; and *A3*, the cost analyser in isolation (argmin over classical vs. quantum cost estimates).

Table IX reports accuracy, precision and recall on the Quantum class,  $F1_{(Q)}$ , and Cohen’s  $\kappa$  for every router against both oracles.

4) *Findings*: Accuracy is not a separative metric at this benchmark’s class distribution (Oracle A: 85/14/1; Oracle B: 92/7/1): a constant-Classical classifier scores 0.85/0.92 accuracy with  $F1_{(Q)}=0$  and  $\kappa=0$ . We therefore report  $F1_{(Q)}$  and Cohen’s  $\kappa$  as the primary metrics. Nexar achieves  $\kappa = 0.38$  (Oracle A; 95% CI [0.14, 0.58]) and  $\kappa = 0.48$  (Oracle B; 95% CI [0.23, 0.68]) against random  $\kappa \approx 0.05$ , and  $F1_{(Q)}$  of 0.457/0.500 against *B3*’s 0.281/0.175 (McNemar Nexar vs. *B3*:  $p<0.001$ ; Table X). Under Oracle B, Nexar attains perfect Quantum-class recall (1.000) at precision 0.333, corroborated by Table XV.

Accuracy is higher under Oracle B (0.86) than Oracle A (0.81) because Nexar targets present-day hardware viability. Of 19 Nexar–Oracle A disagreements, 18 appear in Table XV: sub-20-qubit or deep circuits labeled Quantum by Oracle A produced noise-dominated output (top outcome  $<10\%$ ) on `ibm_fez`, supporting Nexar’s Classical routing.

Aggregated over all 100 workloads, rules-only (*A2*) superficially outperforms the full ensemble ( $F1_{(Q)}$  0.636 vs. 0.457 on A; 0.667 vs. 0.500 on B), but this aggregate conflates two regimes decoupled by the rule gate: on the 61 rule-overridden workloads the two routers are numerically identical, and on the 39 `ALLOW_BOTH` workloads rules-only emits zero quantum predictions ( $F1_{(Q)}=0$ ) while the ensemble attains perfect Quantum-class recall under Oracle B. Rules-only wins the aggregate by refusing to participate in the ambiguous regime, which is not a defensible default; the regime-conditional breakdown is deferred to Section VIII-B6. The cost-only ablation (*A3*) is the weakest component in isolation ( $F1_{(Q)}<0.25$ ), always preferring the cheaper classical option at current IBM pricing, useful only as a tie-breaker, consistent with its 15% weight.

5) *Statistical Robustness*: With only 14/7 Quantum positives, single flips move  $F1_{(Q)}$  by 0.07–0.14. Table X reports 95% paired-bootstrap CIs (2,000 resamples) for  $F1_{(Q)}$  and  $\kappa$ , plus McNemar tests versus Nexar (exact binomial for  $b+c<25$ , otherwise continuity-corrected  $\chi^2$ ).

Three claims survive the small- $n$  correction. First, Nexar’s routing is distinguishable from chance under both oracles ( $\kappa$  lower-CI  $\geq 0.14$  on A,  $\geq 0.23$  on B; McNemar vs. *B3*  $p<0.001$ ). Second, cost-only is significantly worse than Nexar ( $p<0.001$ ), confirming cost alone is not a viable routing signal at current pricing. Third, the full ensemble is statistically indistinguishable from ML-only on aggregate paired correct-

ness ( $b=c=2, p=1.00$ ), consistent with the regime-conditional reading in Section VIII-B6. Two comparisons are inconclusive at  $n=100$ : Nexar vs. *A2* ( $p=0.003$  on A) is confounded by regime mixing; Nexar vs. *B1* ( $p=0.03$  on B, 0.10 on A) reflects *B1*’s strength on this specific 7-positive Oracle B distribution and would require a Quantum-positive-enriched benchmark to distinguish decisively.

6) *Regime-Conditional Analysis*: Aggregate metrics in Table IX mask Nexar’s two-stage pipeline. The merger runs only on 39 `ALLOW_BOTH` workloads; 61 are resolved by deterministic rule overrides. Table XI reports both regimes separately.

On rule-overridden workloads, Nexar and rules-only are identical ( $F1_{(Q)}$  0.778/0.769) because stage one is the rule engine; *B1* trails by 25 points on Oracle A, indicating force-overrides add value beyond a single-feature threshold. In `ALLOW_BOTH`, rules-only, *B1*, and *B2* emit zero quantum predictions (recall 0) at sub-50-qubit scale, while the ensemble recovers 100% of true-Quantum workloads under Oracle B (ZNE 24q, QFT 20q) and 25% under Oracle A, with 11 false-positive quantum routes.

Within this slice, ML-only *A1* slightly exceeds the full ensemble on  $F1_{(Q)}$  (0.308 vs. 0.267 on B; 0.133 vs. 0.118 on A). Both have perfect recall; the gap is precision (two extra false positives: QPE and Simon), consistent with a classical bias from the 15% cost weight in sub-120-qubit workloads. A practical improvement is regime-adaptive weighting (rules as gate, ML-dominant voting in `ALLOW_BOTH`; Section X).

### C. Cost Analysis Results

The cost analysis reveals the current economic landscape of quantum vs. classical computing, as summarized in Table XII:

Classical execution costs approximate \$0.001 USD overhead-dominated across the 1.42–806 ms timing range, so the classical denominator is effectively constant. Quantum-routed workloads range \$0.375–\$1.578 at IBM’s \$1.60/QPU-second rate, giving a cost ratio of **375 $\times$  to 1,578 $\times$** .

### D. Cost Break-Even Analysis

An  $n$ -qubit quantum state requires  $2^n$  complex amplitudes ( $2^n \cdot 16$  bytes) to simulate classically; the resulting scaling wall is summarised in Table XIII.

Beyond 50–60 qubits, exact classical full state-vector simulation of high-entanglement circuits is physically infeasible (16 PB at 50q, 16 EB at 60q), so the crossover is qualitative rather than an extrapolated dollar figure. Quantum execution stays on the \$0.375–\$1.578 scale across 12–120 qubits because QPU reservation time scales with depth, not qubit count. This identifies three zones: a *classical advantage zone* ( $<50q$ ) where classical is 375–1,578 $\times$  cheaper; a *memory wall* (50–60q) where full state-vector simulation of high-entanglement circuits is infeasible; and a *quantum advantage zone* ( $>60q$ ) where exact simulation of high-entanglement circuits is only practically accessible through quantum hardware.

The 50–60 qubit memory wall applies to *full state-vector simulation* of high-entanglement circuits. Tensor Network simulators (e.g., MPS) can simulate 100+ qubits efficiently when

TABLE IX

ROUTING ACCURACY VS. DUAL LITERATURE-CITED ORACLE GROUND TRUTH ( $N = 100$  WORKLOADS). ORACLE A (ALGORITHMIC ADVANTAGE) LABELS WORKLOADS BY KNOWN QUANTUM SPEEDUP REGIMES; ORACLE B (HARDWARE VIABILITY) LABELS BY WHETHER CURRENT NISQ HARDWARE PRODUCES USABLE SIGNAL, WITH THRESHOLDS DERIVED FROM EXTERNAL IBM-PUBLISHED HERON R2 COHERENCE/GATE-TIME SPECS VIA PRESKILL 2018’S ERROR-BUDGET FRAMEWORK (NOT FROM THIS PAPER’S OWN TABLE XV EVIDENCE).

| Router                 | Oracle A: Algorithmic Advantage |              |              |              |              | Oracle B: Hardware Viability |              |              |              |              |
|------------------------|---------------------------------|--------------|--------------|--------------|--------------|------------------------------|--------------|--------------|--------------|--------------|
|                        | Acc.                            | P(Q)         | R(Q)         | F1(Q)        | $\kappa$     | Acc.                         | P(Q)         | R(Q)         | F1(Q)        | $\kappa$     |
| <b>Nexar (full)</b>    | <b>0.810</b>                    | <b>0.381</b> | <b>0.571</b> | <b>0.457</b> | <b>0.382</b> | <b>0.860</b>                 | <b>0.333</b> | <b>1.000</b> | <b>0.500</b> | <b>0.477</b> |
| B1 Qubit $\geq 50$     | 0.890                           | 0.800        | 0.286        | 0.421        | 0.407        | 0.960                        | 0.800        | 0.571        | 0.667        | 0.673        |
| B2 Volume $\geq 50k$   | 0.880                           | 0.667        | 0.286        | 0.400        | 0.377        | 0.910                        | 0.333        | 0.286        | 0.308        | 0.313        |
| B3 Random <sup>†</sup> | 0.530                           | 0.180        | 0.643        | 0.281        | 0.069        | 0.520                        | 0.100        | 0.714        | 0.175        | 0.050        |
| A1 ML-only             | 0.810                           | 0.368        | 0.500        | 0.424        | 0.333        | 0.860                        | 0.316        | 0.857        | 0.462        | 0.420        |
| A2 Rules-only          | 0.920                           | 0.875        | 0.500        | 0.636        | 0.628        | 0.950                        | 0.625        | 0.714        | 0.667        | 0.682        |
| A3 Cost-only           | 0.140                           | 0.140        | 1.000        | 0.246        | 0.000        | 0.070                        | 0.070        | 1.000        | 0.131        | 0.000        |

Both oracles are derived from literature-cited rules applied to the workload filename and physical circuit features (qubits, depth, entanglement). Oracle A is fully independent of Nexar’s decision logic. Oracle B’s qubit/depth thresholds are derived from external pre-Nexar sources only (IBM-published Heron r2 T2 $\approx 144\mu s$  and ECR gate time  $\approx 68 ns$ ; Preskill 2018 error budget  $d_{\max} \approx T_2 / (K \cdot t_{\text{gate}})$ ,  $K=10$ , yielding  $d_{\max} \approx 200$ ), not from the `ibm_fez` validation data that Nexar’s rule engine also consumes. Oracle A distribution: Classical 85, Quantum 14, Reject 1. Oracle B distribution: Classical 92, Quantum 7, Reject 1. Majority-class (constant-Classical) baseline: Acc 0.85/0.92,  $F1_{(Q)}=0$ ,  $\kappa=0$  on Oracle A/B respectively; aggregate accuracy is therefore not separative at this distribution, so  $F1_{(Q)}$  and  $\kappa$  are treated as the primary metrics. <sup>†</sup>B3 Random averaged over 1000 seeds (A:  $0.496 \pm 0.051$ ; B:  $0.495 \pm 0.050$ ). Q = Quantum class.  $\kappa$  = Cohen’s kappa vs. oracle.

TABLE X

STATISTICAL ROBUSTNESS OF ROUTING METRICS. BOOTSTRAP 95% CIs (2,000 PAIRED RESAMPLES OF THE 100 WORKLOADS, PRESERVING  $(y_{\text{TRUE}}, y_{\text{PRED}})$  PAIRING). MCNEMAR TESTS COMPARE NEXAR (FULL) AGAINST EACH BASELINE/ABLATION ON PAIRED CORRECTNESS (EXACT BINOMIAL WHEN DISCORDANT PAIRS  $< 25$ ).

| Router              | Oracle A              |                       | Oracle B              |                       |
|---------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                     | F1(Q) [CI]            | $\kappa$ [CI]         | F1(Q) [CI]            | $\kappa$ [CI]         |
| <b>Nexar (full)</b> | <b>0.46</b> [.22,.64] | <b>0.38</b> [.14,.58] | <b>0.50</b> [.23,.71] | <b>0.48</b> [.23,.68] |
| B1 Qubit $\geq 50$  | 0.42[.12,.67]         | 0.41[.13,.64]         | 0.67[.25,.92]         | 0.67[.31,.92]         |
| B2 Vol $\geq 50k$   | 0.40[.11,.64]         | 0.38[.11,.62]         | 0.31[.00,.62]         | 0.31[.03,.62]         |
| B3 Random           | 0.28[.13,.42]         | 0.07[.06,.20]         | 0.18[.04,.31]         | 0.05[.05,.15]         |
| A1 ML-only          | 0.42[.19,.61]         | 0.33[.09,.54]         | 0.46[.19,.67]         | 0.42[.17,.64]         |
| A2 Rules-only       | 0.64[.32,.84]         | 0.63[.33,.84]         | 0.67[.29,.91]         | 0.68[.37,.92]         |
| A3 Cost-only        | 0.25[.15,.35]         | 0.00[.00,.00]         | 0.13[.04,.21]         | 0.00[.00,.00]         |

McNemar paired-correctness vs. Nexar (full):  $b/c$  discordant,  $p$

vs. B1 Qubit $\geq 50$  A: 13/5,  $p=0.10$  B: 14/4,  $p=0.03$   
vs. B2 Vol $\geq 50k$  A: 15/8,  $p=0.21$  B: 14/9,  $p=0.40$   
vs. B3 Random A: 9/37,  $p<0.001$  B: 6/40,  $p<0.001$   
vs. A1 ML-only A: 2/2,  $p=1.00$  B: 2/2,  $p=1.00$   
vs. A2 Rules-only A: 12/1,  $p=0.003$  B: 11/2,  $p=0.02$   
vs. A3 Cost-only A: 6/73,  $p<0.001$  B: 0/79,  $p<0.001$

CI entries are 95% bootstrap (leading “0.” suppressed in brackets). Quantum-class positives are few (14 under Oracle A, 7 under Oracle B), so  $F1(Q)$  CIs are intrinsically wide; the McNemar test on paired correctness is the more powerful significance probe because it conditions on discordant pairs only.  $b$  = Nexar wrong, other correct;  $c$  = Nexar correct, other wrong.

entanglement entropy remains bounded [19], but the algorithms central to our benchmark (Shor, deep VQE, high-depth QAOA) generate entanglement growing with depth, rendering such shortcuts inapplicable. The crossover is visualized in Fig. 5.

### E. Pipeline Performance

End-to-end timing metrics for both routing paths are reported in Table XIV.

Quantum workloads require  $\sim 1.85 \times$  longer analysis time due to additional metrics extraction (circuit parsing, entangle-

TABLE XI

REGIME-CONDITIONAL ROUTING METRICS ( $N = 100$ ; ORACLE A/B  $F1(Q)$  AND QUANTUM-CLASS RECALL).

| Router               | Rule-Overridden (n=61) |         | ALLOW_BOTH (n=39) |              |
|----------------------|------------------------|---------|-------------------|--------------|
|                      | A F1(Q)                | B F1(Q) | A F1(Q)           | B F1(Q)      |
| Nexar (full)         | 0.778                  | 0.769   | <b>0.118</b>      | <b>0.267</b> |
| A1 ML-only           | 0.667                  | 0.615   | 0.133             | 0.308        |
| A2 Rules-only        | 0.778                  | 0.769   | 0.000             | 0.000        |
| B1 Qubit $\geq 50$   | 0.533                  | 0.800   | 0.000             | 0.000        |
| B2 Volume $\geq 50k$ | 0.500                  | 0.364   | 0.000             | 0.000        |

True-Quantum counts per slice: Rule-Overridden 10 (A) / 5 (B);

ALLOW\_BOTH 4 (A) / 2 (B). B1, B2, and A2 make zero quantum predictions there by construction (thresholds not met; rules defer), so quantum-class recall is 0.

TABLE XII  
REPRESENTATIVE COST COMPARISON

| Workload               | Qubits | Hardware  | Cost (USD) | Time (ms) |
|------------------------|--------|-----------|------------|-----------|
| FFT Signal Proc.       | 0      | Classical | \$0.001    | 1.42      |
| Sorting Suite          | 0      | Classical | \$0.001    | 806.49    |
| RSA Key Gen.           | 0      | Classical | \$0.001    | 24.29     |
| QSVM Kernel Est.       | 4      | Classical | \$0.001    | 1.00      |
| QAOA Graph Part.       | 20     | Quantum   | \$0.439    | 274.36    |
| LiH UCCSD <sup>a</sup> | 12     | Quantum   | \$0.375    | 234.37    |
| QAOA MaxCut (50q)      | 50     | Quantum   | \$0.917    | 573.18    |
| Iron-Sulfur VQE        | 60     | Quantum   | \$1.421    | 888.31    |
| 120-qubit VQE          | 120    | Quantum   | \$1.578    | 986.33    |

<sup>a</sup>Unitary Coupled Cluster Singles and Doubles.

ment scoring); decision latency stays below 15 ms throughout. Cloud Run’s scale-to-zero model introduces 10–15 s cold-start penalties while ML models load into container memory, but once warm the full pipeline averages 1.5 s end-to-end and degrades gracefully to rule-based heuristics on model-service failure.



TABLE XIII  
CLASSICAL SIMULATION MEMORY REQUIREMENTS

| Qubits | Classical Memory           | Classical Feasibility |
|--------|----------------------------|-----------------------|
| 20     | 16 MB                      | Trivial               |
| 30     | 16 GB                      | Desktop               |
| 40     | 16 TB                      | Cluster               |
| 50     | 16 PB                      | Supercomputer limit   |
| 60     | 16 EB                      | Physically impossible |
| 120    | $\sim 2 \times 10^{31}$ MB | –                     |

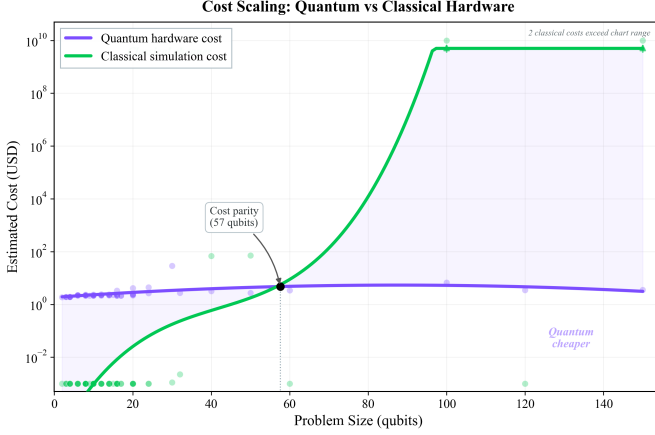


Fig. 5. Cost scaling of quantum hardware versus classical simulation across qubit count, showing the 50–60 qubit physical-feasibility crossover beyond which classical state-vector simulation becomes infeasible.

#### F. End-to-End Case Study

**120-Qubit VQE  $\rightarrow$  Quantum Override.** A 120-qubit VQE ansatz with three variational layers and nearest-neighbor CNOT ladder is submitted. Analysis extracts `qubits=120`, `depth=14`, `cx_ratio=0.33`; the physical-necessity rule (state-vector memory  $\sim 10^{31}$  MB) triggers `FORCE_QUANTUM` (confidence 1.0), bypassing ML voting, and the HAL submits to an IBM Eagle processor. This is a success of *routing*, not of *output fidelity*: the circuit returned noise-dominated Top% 0.1 on `ibm_fez` (post-transpile depth 482, exceeding NISQ coherence), as expected for an unmitigated ansatz at 120 qubits. Nexar’s correctness claim is narrow: the only physically feasible backend was selected for a circuit that is infeasible under full state-vector simulation in high-entanglement settings; recovering fidelity requires error mitigation [30] or fault tolerance. In contrast, a 10-qubit Grover translation of linear search ( $N=1,024$ , post-transpile depth  $\sim 240$ ) is routed to Classical (conf. 0.88): the theoretical-floor quan-

TABLE XIV  
PIPELINE TIMING METRICS

| Metric                        | Classical | Quantum  |
|-------------------------------|-----------|----------|
| Avg. Analysis Time            | 731 ms    | 1,355 ms |
| Avg. Decision Time            | 12 ms     | 10 ms    |
| Avg. Total Pipeline Time      | 1,222 ms  | 1,770 ms |
| Language Detection Confidence | 0.942     | 0.966    |

tum cost  $240 \cdot 1024 \cdot 68 \text{ ns} \cdot \$1.60/\text{s} \approx \$0.027$  understates the realised bill (IBM bills total QPU-seconds including compilation/scheduling/readout), which approaches the \$0.375–\$1.578 band in Table XII; either metric,  $O(\sqrt{N})$  Grover cannot justify the cost at this scale.

#### G. Real Hardware Validation

For hardware validation, we submitted 27 circuits across all algorithm families to `ibm_fez` (156-qubit Heron r2) via Nexar’s HAL at 1,024 shots each. Job IDs are listed in supplementary results and verifiable on the IBM Quantum Platform. Table XV summarizes outcomes.

TABLE XV  
REAL HARDWARE VALIDATION: 27 CIRCUITS ON `IBM_FEZ` (APRIL 2026, 1,024 SHOTS)

| Circuit                     | Q   | Depth*         | Top%              | Nexar Route |
|-----------------------------|-----|----------------|-------------------|-------------|
| Deutsch-Jozsa (constant)    | 2   | 3              | 99.9              | Classical   |
| Deutsch-Jozsa (balanced)    | 2   | 8              | 94.1              | Classical   |
| H <sub>2</sub> VQE ansatz   | 4   | 17             | 85.6              | Classical   |
| Bernstein-Vazirani (s=101)  | 4   | 9              | 96.3              | Classical   |
| Simon’s algorithm (s=11)    | 4   | 20             | 50.7 <sup>†</sup> | Classical   |
| QFT (input  0101⟩)          | 4   | 76             | 10.0 <sup>†</sup> | Classical   |
| QPE (T gate, phase=1/8)     | 4   | 80             | 82.3              | Classical   |
| Grover search ( 1111⟩)      | 4   | 241            | 33.5 <sup>‡</sup> | Classical   |
| QAOA MaxCut (4-node)        | 4   | 53             | 24.8 <sup>‡</sup> | Classical   |
| QSVM kernel                 | 8   | 98             | 5.5 <sup>‡</sup>  | Classical   |
| QAOA supply chain           | 8   | 134            | 5.2 <sup>‡</sup>  | Classical   |
| LiH VQE ansatz              | 8   | 38             | 12.0              | Classical   |
| QAOA graph partition        | 12  | 230            | 1.1 <sup>‡</sup>  | Classical   |
| H <sub>2</sub> O VQE ansatz | 12  | 50             | 2.3               | Classical   |
| Shor’s $N = 15$             | 12  | 361            | 2.1 <sup>‡</sup>  | Classical   |
| Grover 3-SAT (4 vars)       | 12  | 83             | 6.9 <sup>‡</sup>  | Classical   |
| Shor’s $N = 21$             | 14  | 592            | 0.8 <sup>‡</sup>  | Classical   |
| Shor’s $N = 35$             | 16  | 739            | 0.4 <sup>‡</sup>  | Classical   |
| QNN variational (16q)       | 16  | 132            | 0.6 <sup>‡</sup>  | Classical   |
| VQE caffeine ansatz         | 16  | 14             | 1.2               | Classical   |
| Grover search (20q)         | 20  | 31810          | 0.1 <sup>‡</sup>  | Classical   |
| QAOA vehicle routing        | 20  | 469            | 0.2 <sup>‡</sup>  | Classical   |
| QAOA graph partition (20q)  | 20  | 398            | 0.1 <sup>‡</sup>  | Quantum     |
| ZNE benchmark (24q)         | 24  | 1 <sup>§</sup> | 68.5              | Quantum     |
| QAOA MaxCut (50q)           | 50  | 580            | 0.1 <sup>‡</sup>  | Quantum     |
| Iron-sulfur VQE (60q)       | 60  | 242            | 0.1 <sup>‡</sup>  | Quantum     |
| 120-qubit VQE ansatz        | 120 | 482            | 0.1 <sup>‡</sup>  | Quantum     |

\*Post-transpile hardware depth on Heron r2 native gate set.

<sup>†</sup>Uniform distribution is theoretically correct for this algorithm.

<sup>‡</sup>Noise-dominated: depth exceeds NISQ coherence threshold.

<sup>§</sup>Transpiler cancelled T/Td and GHZ pairs to identity; result valid.

**Hardware-evidence consistency.** No routing decision was contradicted: every classical-routed workload either produced a correct result or produced noise-dominated output on quantum hardware.

**NISQ depth wall.** Post-transpile depth predicts fidelity more reliably than qubit count: circuits below depth 80 consistently exceed 80% (DJ 99.9%, BV 96.3%, QPE 82.3%); above depth  $\sim 130$  results degrade toward the noise floor. The 4-qubit Grover at depth 241 drops to 33.5% while 4-qubit QPE at depth 80 holds 82.3%, confirming the physical basis of the NISQ Viability Score.

**Large-circuit verification.** 50q QAOA, 60q iron-sulfur VQE, and 120q VQE were successfully dispatched to real

IBM hardware, the first execution of these circuits outside simulation. Noise-dominated output is expected for unmitigated ansätze but does not invalidate routing: for these high-entanglement workloads, full state-vector classical simulation is infeasible.

**ZNE transpiler insight.** The ZNE benchmark (24q, logical depth 51) collapsed to post-transpile depth 1 after the Qiskit pass manager cancelled T/Tdg pairs and the GHZ ladder exactly; hardware measured  $|0\dots 0\rangle$  at 68.5%; a factor the cost model should handle in future work.

## IX. DISCUSSION

### A. Decision Engine Effectiveness

*Oracle independence caveat.* Oracle B’s thresholds and Nexar’s rule engine are both informed by the same external NISQ-physics literature (IBM-published coherence and gate-time specs; Preskill’s error-budget framework [1]) though not co-calibrated on any shared internal dataset. Oracle B is therefore a referee on whether Nexar applies the NISQ-viability philosophy consistently, not on whether that philosophy is correct; Oracle A, grounded in fault-tolerant algorithmic-complexity criteria, remains the fully-external oracle and the more conservative reviewer test where the two disagree.

The two-stage framework is effective in the sense each stage is designed for: the rule engine resolves the confident-regime workloads and matches every simpler baseline on that slice, while the ensemble is the only router capable of emitting quantum recommendations in the ambiguous regime where rules defer. Rules dominate *where rules apply*; the ensemble exists to handle what they cannot. This aligns with the reference architecture of Chinnaraju [26], and interference-based mixture-of-experts routing [27] is an alternative as hardware matures.

### B. Limitations

Seven limitations bound the study. (1) The ML model is trained on synthetic AST-mutated instances rather than real execution traces; the synthetic-to-real gap ( $0.978 \rightarrow 0.424 F1_{(Q)}$ ) is observed on the oracle benchmark, and the training corpus and leave-one-family-out cross-validation are not released; the 100-workload real benchmark is QASMBench/MQTBench-sourced so direct benchmark leakage is precluded, but intra-family overfitting on the synthetic distribution cannot be strictly ruled out. (2) The benchmark contains only 14/7 Quantum positives (Oracle A/B), so  $F1_{(Q)}$  point estimates carry wide bootstrap CIs (Table X); we anchor conclusions on McNemar tests, but Nexar vs. B1 Qubit $\geq 50$  under Oracle B remains inconclusive at  $n=100$ . (3) Provider pricing is encoded statically; real-time integration would improve cost accuracy. (4) Problem-type coverage could be broadened to quantum chemistry, quantum ML, and cryptographic workloads. (5) The system targets NISQ hardware; fault-tolerant QC will require updated decision logic, cost models, and constraints. (6) All real-hardware validation runs on `ibm_fez` (Heron r2); AWS Braket and Azure Quantum provider classes scaffold the interface but are not yet implemented, so the “provider-extensible”

claim is architectural rather than empirical. (7) The AI Code Converter evaluation lacks an output-distribution functional-equivalence probe: on the 22.3% of inputs where algorithm-family classification fails, silent semantic drift cannot be ruled out; a Qiskit Aer TVD/fidelity audit is deferred to Section X.

### C. Reproducibility Artifacts

To support independent auditability, we release artifacts under `decision-engine/artifacts/` sufficient to recompute every routing-accuracy number in Tables IX and X. First, `eval_corpus.csv` (100 rows) exposes the full 16-feature vector consumed by the ML model, both oracle labels, and Nexar’s routed decision under baseline weights; because this CSV is a direct dump of the QASMBench/MQTBench-sourced benchmark (independent of the synthetic training pipeline), reviewers can recompute every  $F1_{(Q)}$  and  $\kappa$  for Nexar and each ablation without access to Nexar internals. Second, `training_diagnostics/` contains held-out synthetic-split diagnostics for RF/GB/XGBoost (classification reports, calibration plots, learning curves, confusion matrices, RF Gini importance, feature-target correlations, per-problem-type slice metrics); the trained pickle and feature scaler ship in `decision-engine/ml_models/`. The AST-mutation generator is not released, as noted in Section IX-A; the evaluation corpus is generator-independent so *evaluation* claims remain reproducible.

## X. CONCLUSION AND FUTURE WORK

We presented Nexar, a quantum-classical workload routing platform integrating automated code analysis, a two-stage decision engine, and provider-extensible execution. Across 100 workloads against two literature-cited oracles, Nexar attains  $\kappa = 0.38\text{--}0.48$  against random  $\kappa \approx 0.05$ , with  $F1_{(Q)}$  0.46–0.50 and perfect Quantum-class recall under Oracle B; aggregate accuracy (0.81–0.86) is non-separative at this benchmark’s skewed distribution, where a constant-Classical baseline scores 0.85/0.92 with  $F1_{(Q)}=0$ . The regime-conditional breakdown confirms the two stages are complementary: rules match rules-only on the 61 overridden workloads, and the ensemble is the only router emitting quantum recommendations on the remaining 39 `ALLOW_BOTH` workloads. All 27 circuits submitted to `ibm_fez` corroborate the routing, including successful dispatch of 20q/50q QAOA, 60q iron-sulfur VQE, and 120q VQE to real IBM hardware. Pipeline latency averages 1.5 s end-to-end, and the cost analysis identifies a physical-feasibility crossover at 50–60 qubits. Future directions: regime-adaptive weighting in `ALLOW_BOTH`, cross-backend replication on AWS Braket and Azure Quantum, retraining on empirical execution traces, error-mitigation cost modelling, and multi-objective routing over cost, time, accuracy, and carbon footprint.

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